PROJECT PROPOSAL

PROJECT GROUP NO:06 **ACADEMIC YEAR: 2025-26**

A. Proposed Title of the Project

Resolving Domain-Specific Limitations in 7B Models via Fine-Tuning and Retrieval-Augmented Generation

B. Introduction

Large Language Models (LLMs) have revolutionized natural language processing (NLP), enabling impressive applications across various domains. However, smaller models, such as those with 7 billion parameters, still struggle with specialized knowledge and real-time information retrieval compared to their much larger counterparts. This creates a challenge for making AI more efficient without requiring massive computational resources.

This project explores ways to enhance the performance of smaller models using two key techniques: **domain-specific fine-tuning** and **retrieval-augmented generation (RAG)**. Fine-tuning helps models adapt better to specific topics, improving their accuracy in those areas. Meanwhile, RAG enhances responses by pulling in relevant external information during inference, making the model more reliable and context-aware.

By applying these methods, this research aims to improve accuracy, relevance, and efficiency while reducing hallucinations in a 7B model. The goal is to bridge the gap between smaller and larger models, making advanced AI capabilities more accessible even in resource-limited environments where massive models aren't practical due to hardware or cost constraints.

C. Literature Review

Large Language Models (LLMs) have advanced NLP tasks like translation and question answering but often require extensive computational resources, limiting accessibility. To overcome this, researchers are focusing on optimizing smaller 7B-parameter models using two key strategies: parameter-efficient fine-tuning and retrieval-augmented generation (RAG).

Parameter-Efficient Fine-Tuning:
Traditional fine-tuning updates all model parameters, which is resource-intensive. Techniques such as **Quantized LoRA (QLoRA)** and **Low-Rank Adaptation (LoRA)** update only a small subset of parameters, drastically reducing memory usage and computational costs. QLoRA quantizes weights (e.g., to 4-bit precision) to boost performance metrics like BLEU scores, while LoRA leverages low-rank matrix updates to maintain efficiency even on limited hardware.

Retrieval-Augmented Generation (RAG):
LLMs can experience hallucinations and knowledge staleness when relying solely on their internal data. RAG addresses these issues by dynamically incorporating external information during inference. Advanced methods like **ITER-RETGEN** iteratively refine the retrieval process—generating an initial answer and then updating the query—to improve factual accuracy by around 8.6% on benchmark tasks.

Combining Fine-Tuning and RAG:
Integrating fine-tuning techniques with RAG methods leverages the strengths of both approaches. For example, when QLoRA or LoRA fine-tuning is combined with RAG—and enhanced further with strategies such as Translation Memory Prompting (TMPLM)—the resulting system can significantly boost performance. This combined approach not only improves metrics like BLEU scores but also enhances the model's ability to handle low-resource languages and dynamic information updates.

Future Directions:
Building on these successes, ongoing research is exploring several promising directions. These include hybrid optimization methods that tightly integrate iterative retrieval with fine-tuning, extending these methods to low-resource languages, and even venturing into multi-modal retrieval to incorporate text, images, and audio. Additionally, real-time retrieval mechanisms are under development to ensure that LLMs can continuously update their knowledge base and further minimize errors.

D. Objectives

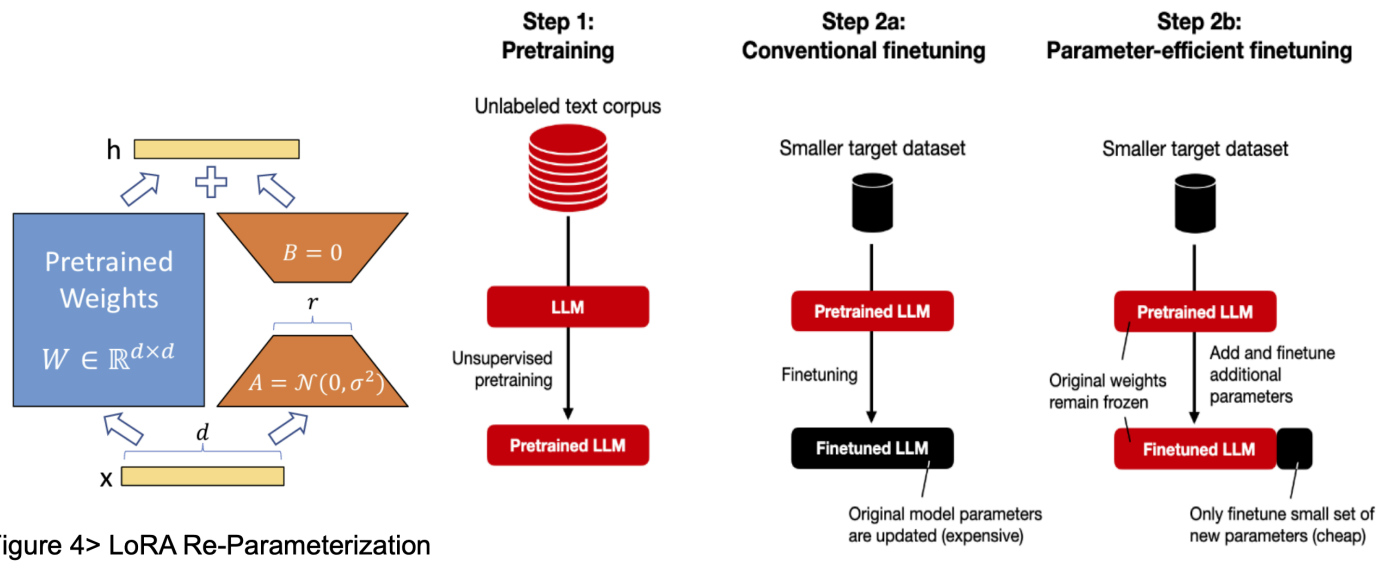
1. To explore and document the specific limitations of an open-source 7B parameter language model (like Llama 2 7B or Mistral 7B) on knowledge-intensive tasks through practical testing.
2. To implement a basic domain-specific fine-tuning procedure on a selected 7B model using a modest dataset in a specific domain (e.g., computer science or healthcare).
3. To build a simple retrieval-augmented generation (RAG) system that connects our 7B model to external knowledge sources through vector databases.
4. To compare the performance of our baseline model against both the fine-tuned version and the RAG-enhanced version across a small set of evaluation tasks.
5. To analyze the computational requirements and practical challenges of implementing these techniques on consumer-grade hardware available to students.

E. Workplace	Address of Other Institute / Industry (if any):
SMVITM,Bantakal	

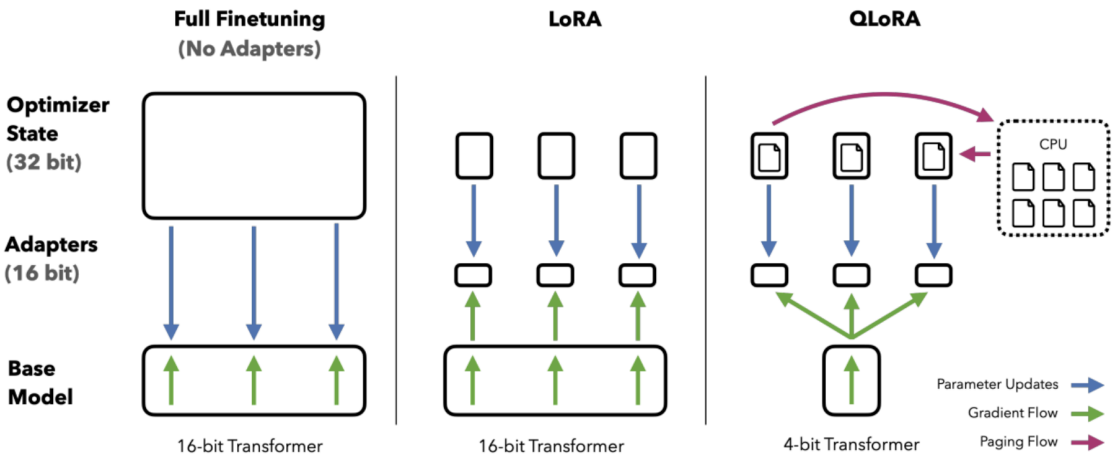
F. Requirement Details

1. Hardware	1. **GPU**: A high-performance GPU (e.g., NVIDIA A100, RTX 3090, or Tesla T4) for efficient model training and inference. 2. **CPU**: Multi-core processor (e.g., Intel i7/i9, AMD Ryzen 7/9) for preprocessing and model execution. 3. **RAM**: At least **32GB** for smooth data handling and training. 4. **Storage**: Minimum **500GB SSD**, preferably **1TB+**, to store datasets, models, and checkpoints. 5. **Cloud Access (Optional)**: Google Colab Pro, AWS EC2, or Azure VM with GPU support if local hardware is insufficient.

2. Software	1. **Operating System**: Windows, Linux (Ubuntu preferred), or macOS. 2. **Programming Language**: Python (with PyTorch or TensorFlow for model training). 3. **Machine Learning Libraries**: ● PyTorch / TensorFlow (for model training and fine-tuning). ● Hugging Face Transformers (for working with LLMs). ● FAISS / ChromaDB (for implementing retrieval-augmented generation). 4. **Data Processing & Analysis**: ● Pandas, NumPy, and Matplotlib for dataset handling and visualization. 5. **Development Tools**: ● Jupyter Notebook / VS Code / PyCharm for coding.
3. Skill sets	1. **Machine Learning & Deep Learning**: Understanding of neural networks, model fine-tuning, and evaluation metrics. 2. **Natural Language Processing (NLP)**: Familiarity with tokenization, embeddings, and transformer architectures. 3. **Python Programming**: Proficiency in Python for implementing ML pipelines. 4. **Data Handling & Preprocessing**: Skills in managing structured and unstructured data. 5. **Model Deployment & Optimization**: Knowledge of how to efficiently run models on limited hardware. 6. **Cloud Computing (Optional)**: Experience with cloud platforms for training and deploying models.
4. Others	1. **Datasets**: Domain-specific datasets for fine-tuning the model. 2. **Pre-trained Models**: Access to 7B parameter models from Hugging Face or similar sources. 3. **Documentation & Research Papers**: Reference materials for implementing fine-tuning and RAG techniques. 4. **Collaboration & Experimentation**: A structured approach to testing different model enhancements.
G. Figure(s)/Drawing(s)	



<Figure 4> LoRA Re-Parameterization



<Figure 7> QLoRA improves over LoRA

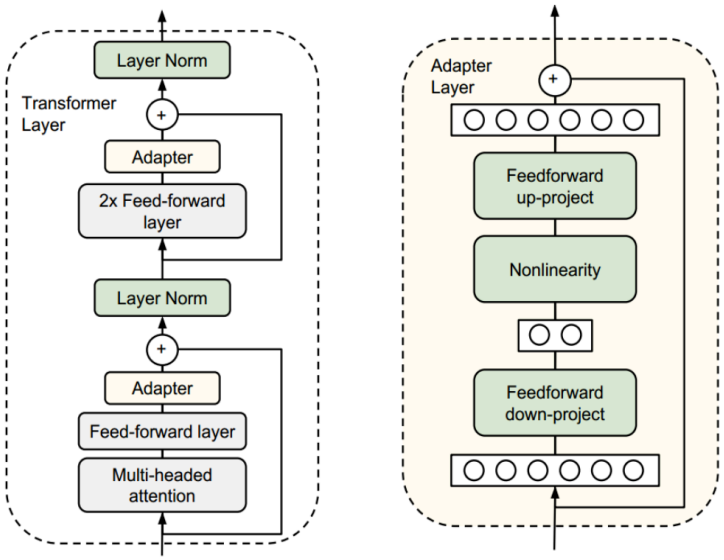
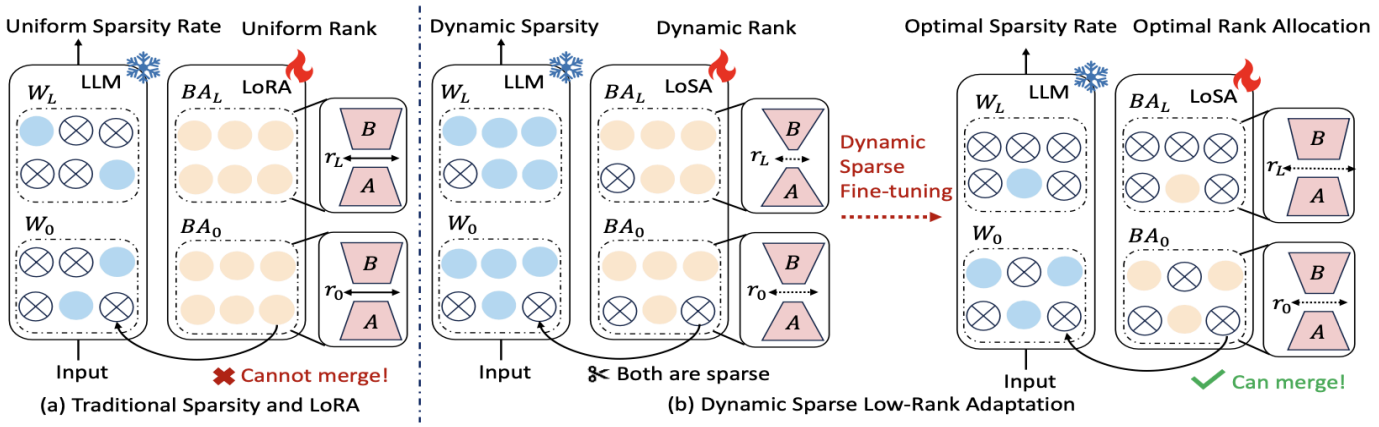


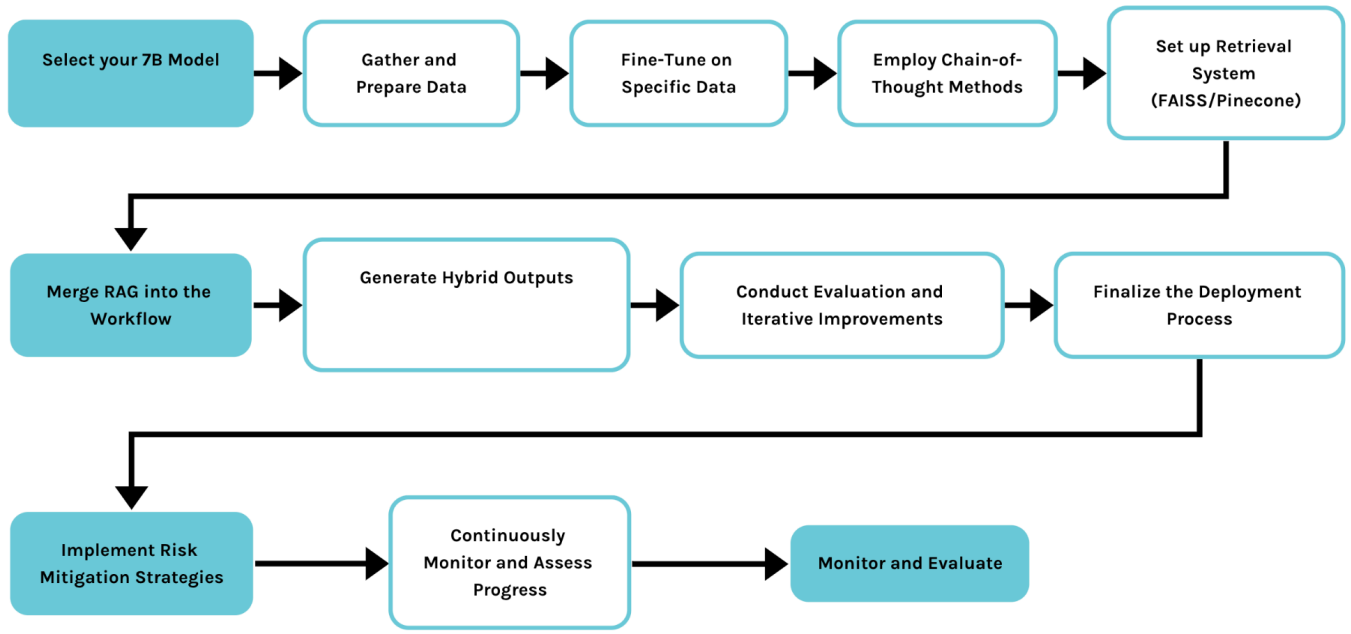
Fig. 3: The basic structure of Adapter Tuning [52].



H. Methodology

1. **Data Collection & Preprocessing:**
   - **Data Curation:** Gather high-quality research datasets and academic materials specific to the target domain.
   - **Annotation:** Incorporate chain-of-thought examples that break down complex reasoning steps.
   - **Preprocessing:** Clean, format, and split the dataset into training, validation, and test sets.
2. **Fine-Tuning the 7B Model:**
   - **Model Adaptation:** Fine-tune the pre-trained 7B model on the curated dataset.
   - **Chain-of-Thought Integration:** Train the model with chain-of-thought prompts to enhance multi-step reasoning.
   - **Parameter-Efficient Techniques:** Utilize methods such as LoRA or adapter modules to fine-tune without losing generalization capabilities.
3. **Retrieval-Augmented Generation (RAG) Integration:**
   - **Retrieval System Setup:** Deploy a vector-based retrieval system (e.g., FAISS or Pinecone) to index external knowledge sources.
   - **Hybrid Pipeline:** Develop a pipeline where the model's output is supplemented by retrieved documents or data.
   - **Response Merging:** Implement mechanisms to merge or refine the responses from both the fine-tuned model and the retrieval component.
4. **Evaluation & Iteration:**
   - **Testing:** Evaluate the system on complex research queries across various metrics (accuracy, depth, contextual relevance).
   - **Feedback Loop:** Refine the model and retrieval integration based on performance evaluations and user feedback.
   - **Iteration:** Continuously update both the fine-tuning and retrieval mechanisms to adapt to evolving research needs.
5. **Deployment:**
   - **Interface Integration:** Develop an application interface or API endpoint to facilitate user interactions.
   - **Scalability:** Ensure that the integrated system supports real-time retrieval and response generation.
   - **Monitoring:** Set up performance monitoring and maintenance routines to update the model and data sources as necessary.

I. Flowchart (Plan of Action)



J. Expected Outcome of the project

1. A fine-tuned 7B model on a specific domain showing measurable improvements over the base model on our test queries.
2. A working RAG prototype that can retrieve relevant information from our curated knowledge base and augment the model's responses.
3. Empirical data comparing the three approaches (base model, fine-tuned, and RAG) on metrics like accuracy, relevance, and hallucination rate for our test cases.
4. Documentation of the technical challenges encountered and solutions implemented throughout the project.
5. A demonstration of the system showcasing example queries where our enhanced model outperforms the baseline model.
6. A project report and presentation detailing our methodology, implementation, results, and lessons earned for future work.

K. Estimated cost	Source of Fund

L. References

1.T. Brown, B. Mann, N. Ryder et al., “Language Models are Few-Shot Learners,” Advances in Neural Information Processing Systems, vol. 33,pp. 1877–1901, 2020.
2.T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, “QLoRA:Efficient Fine-Tuning of Quantized LLMs,” arXiv preprint arXiv:2305.14314, 2023.
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M. Project Batch members

S.No.	USN	Name	Signature with date
1.	4MW22CS015	Akshaykumar	
2.	4MW22CS015	Annson Dsouza	
3.	4MW22CS015	Dhanush S Shet	
4.	4MW22CS015	Justin Aroza	

N. Accepted/Rejected:

O. Suggestions for implementation:

P. Guidance

Guide (s) allotted	Guide Name and Signature (s) with date
1. Guide	

Q. Review Comments/Suggestions	Accept/Modify/ Reject	Reviewer Signature & Name